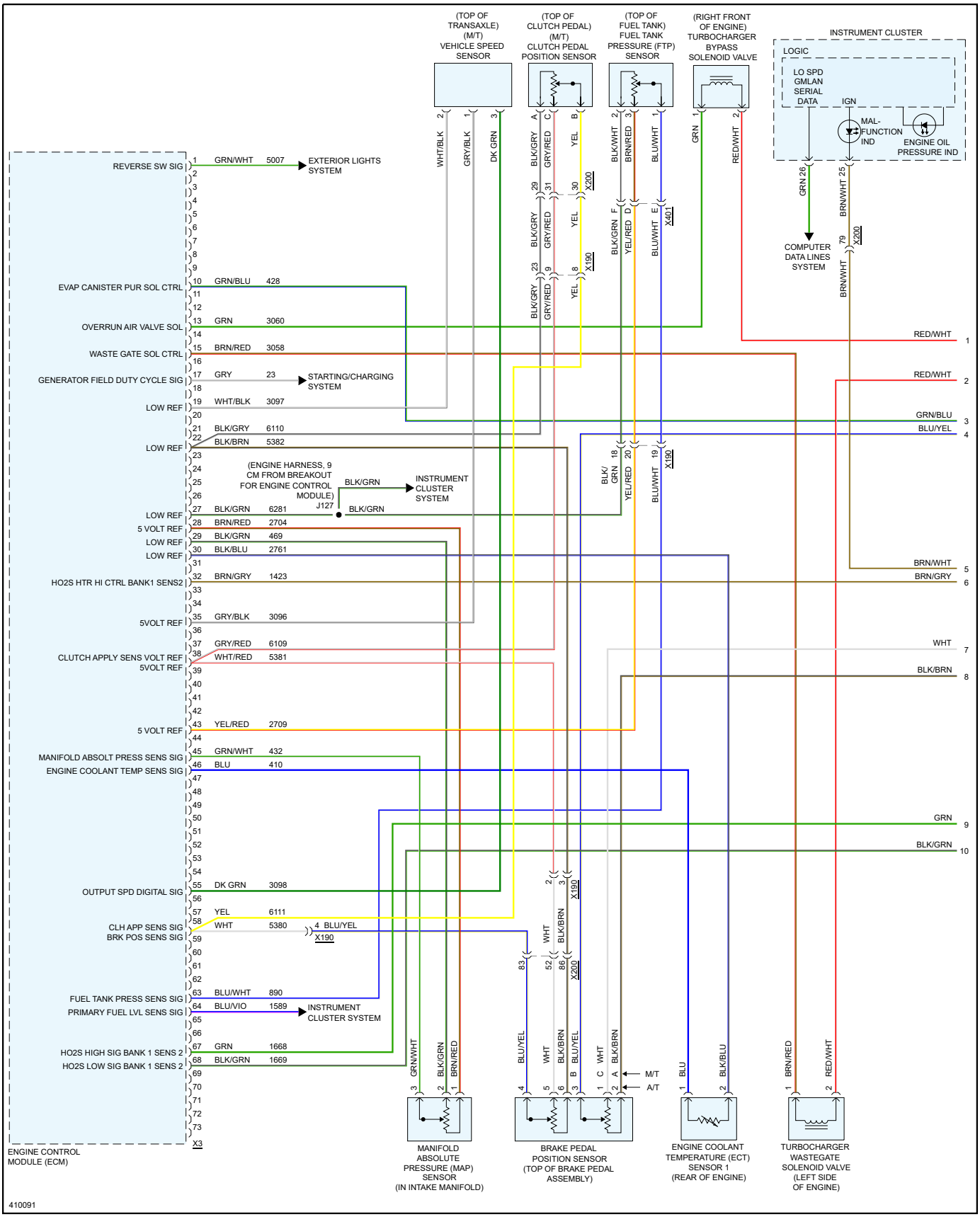
2013 Chevrolet Cruze L4-1.4L Turbo

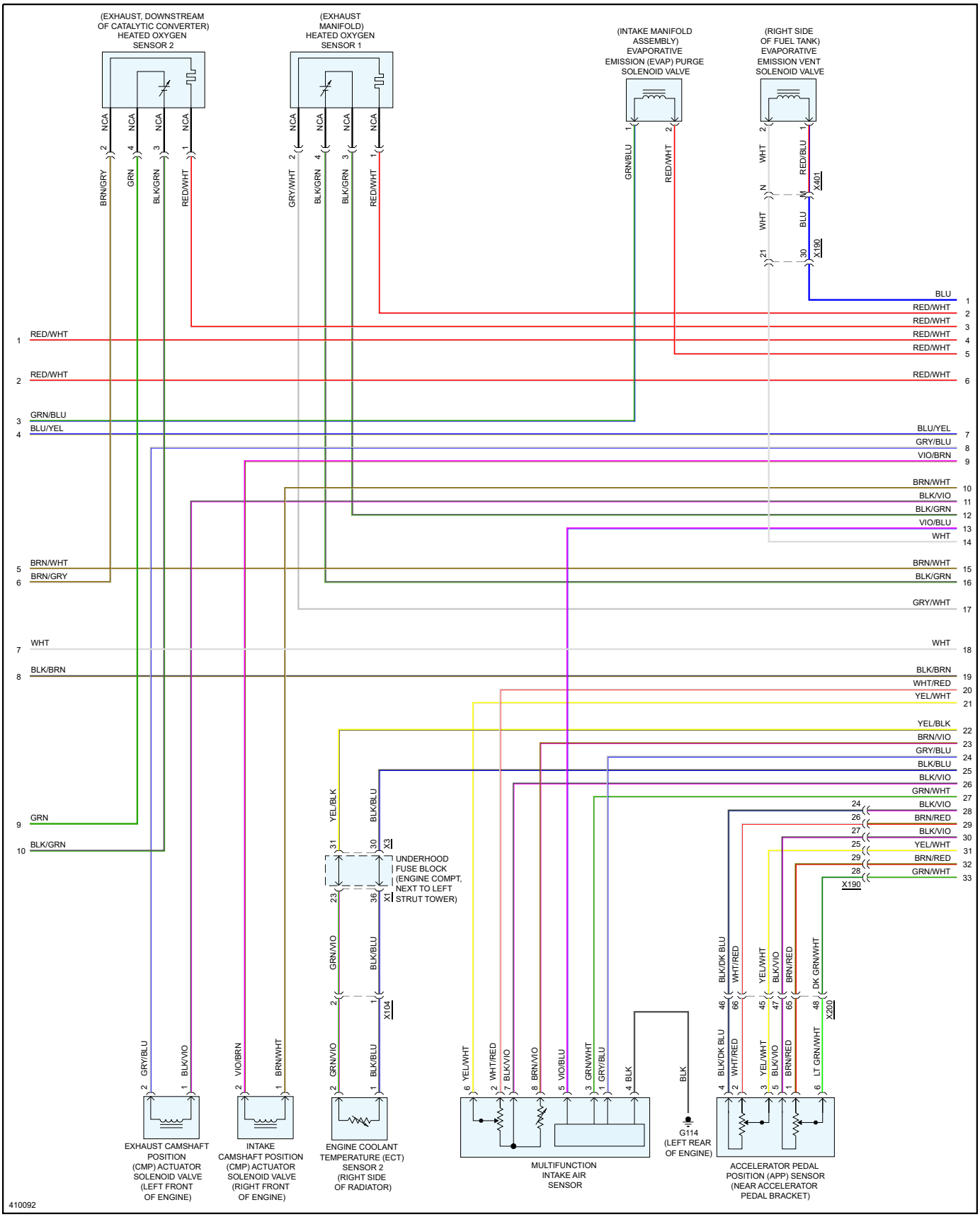
Vehicle > Powertrain Management > Diagrams > Electrical - Interactive Color (Non OE)

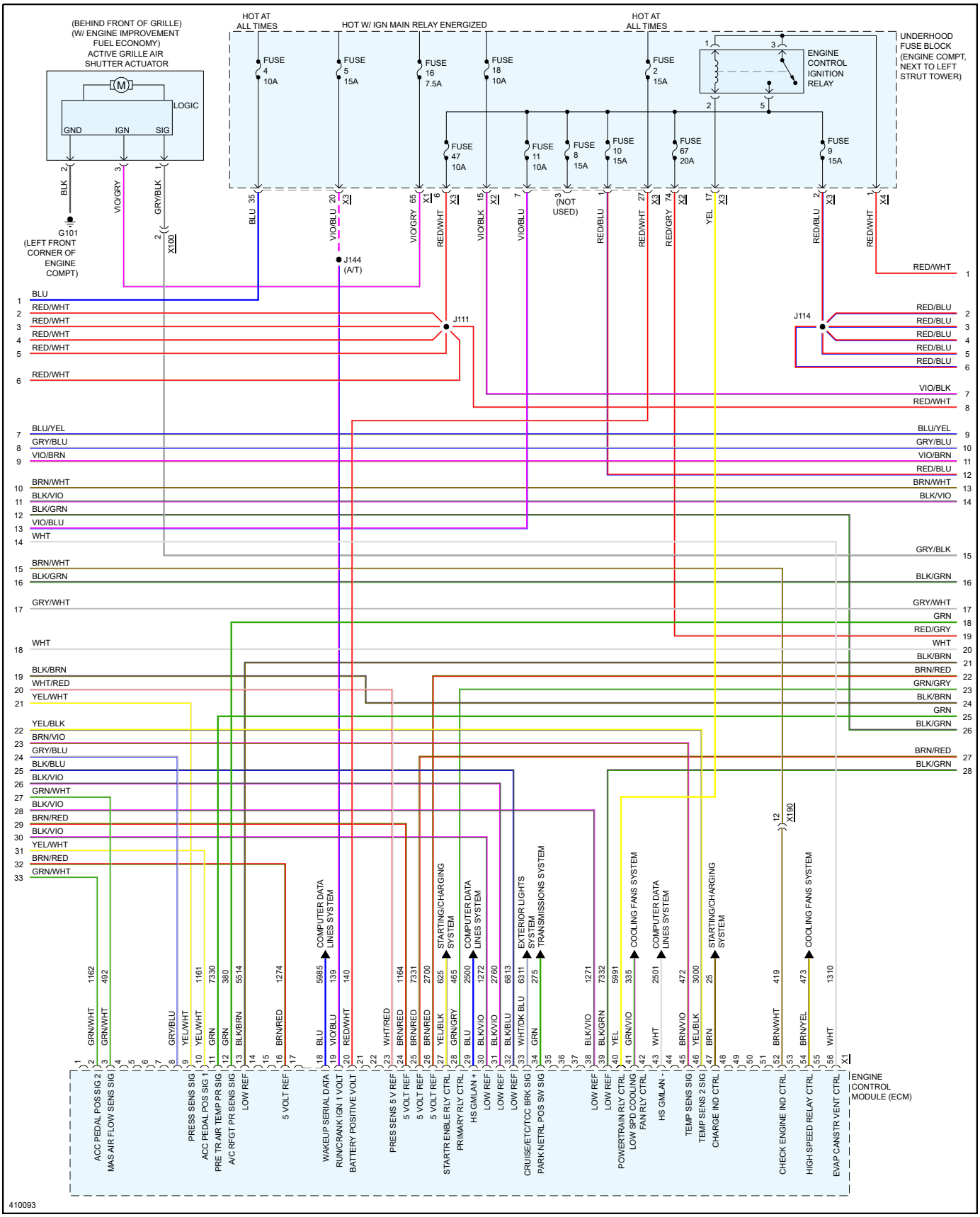
ENGINE CONTROLS

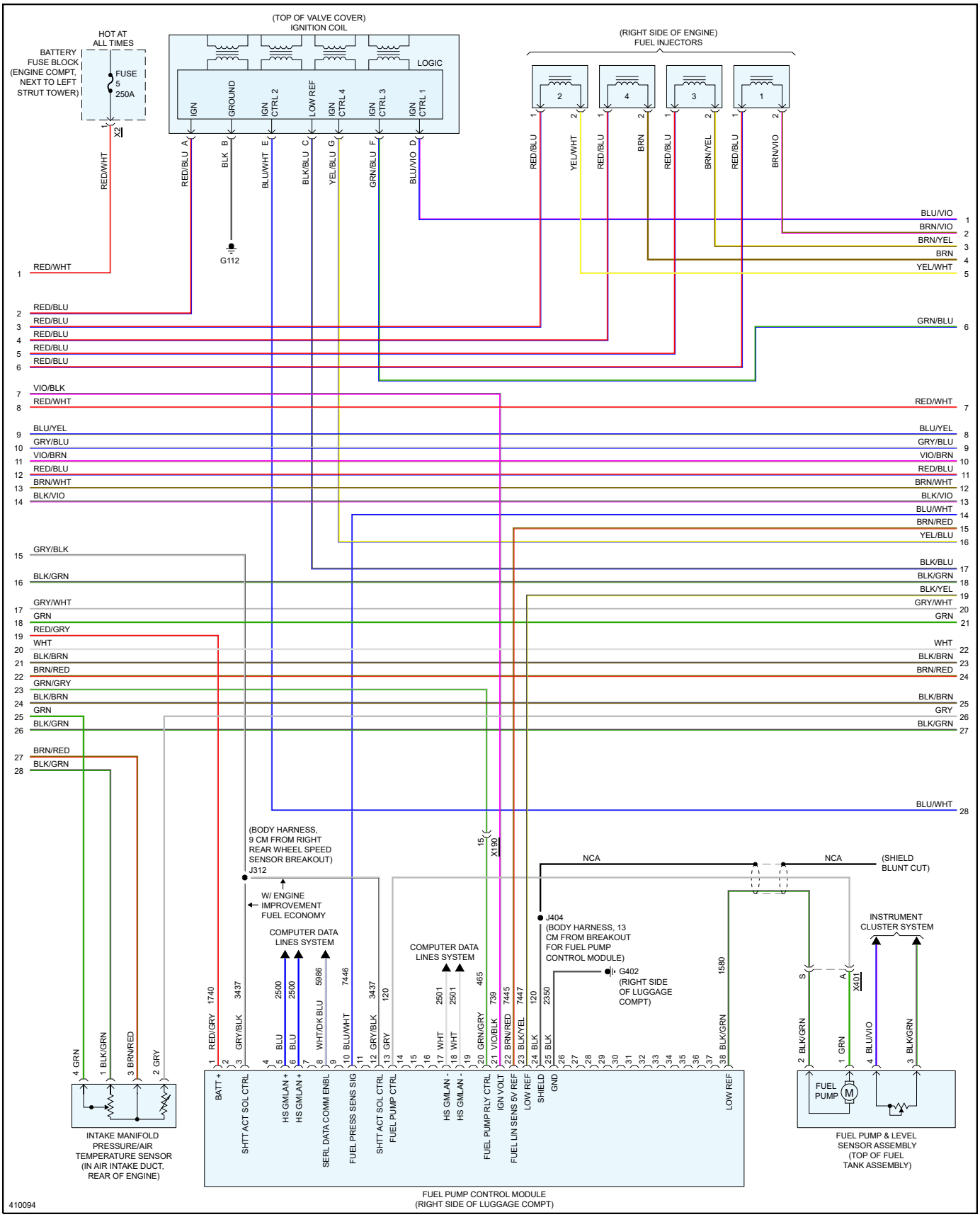
Engine Controls

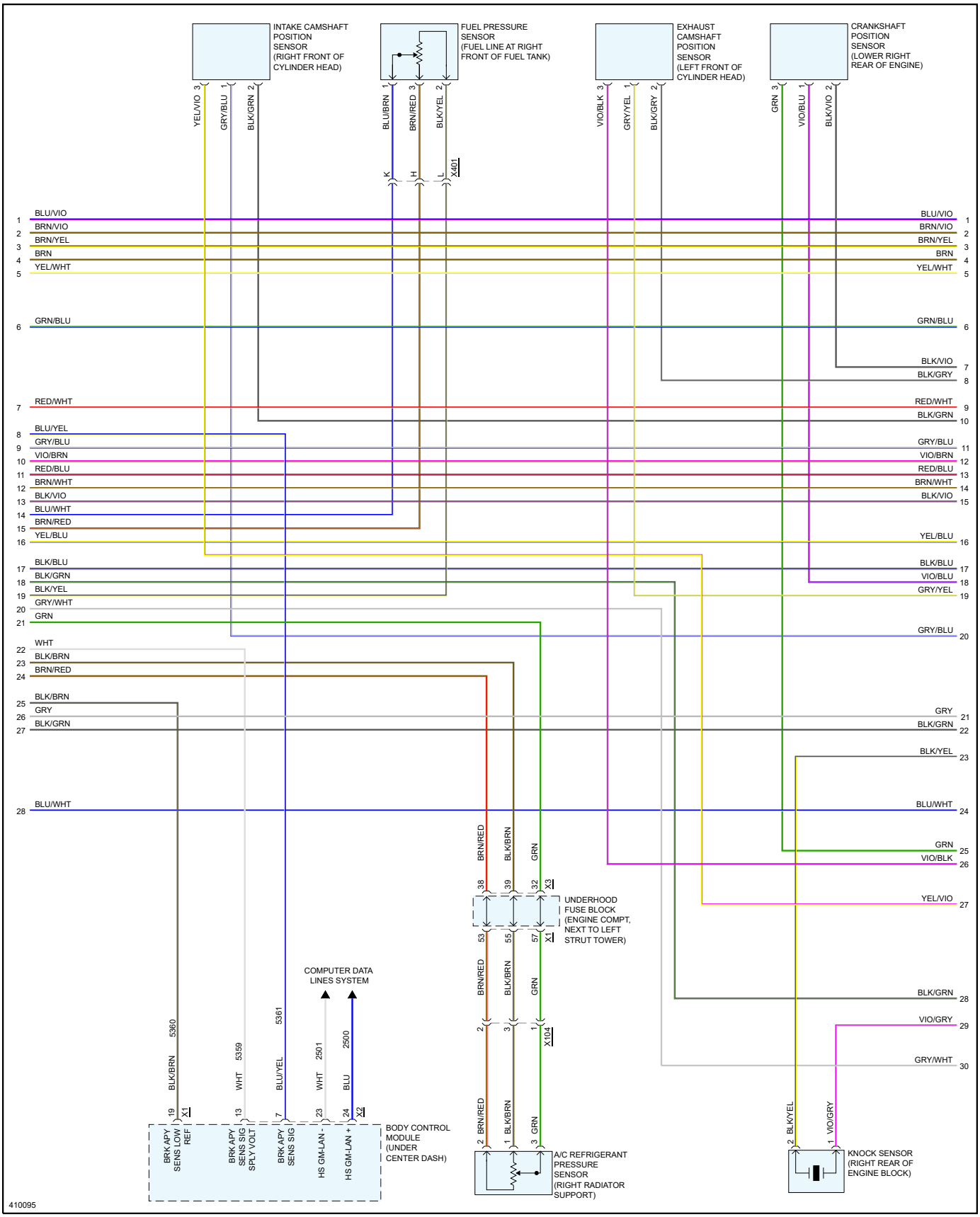
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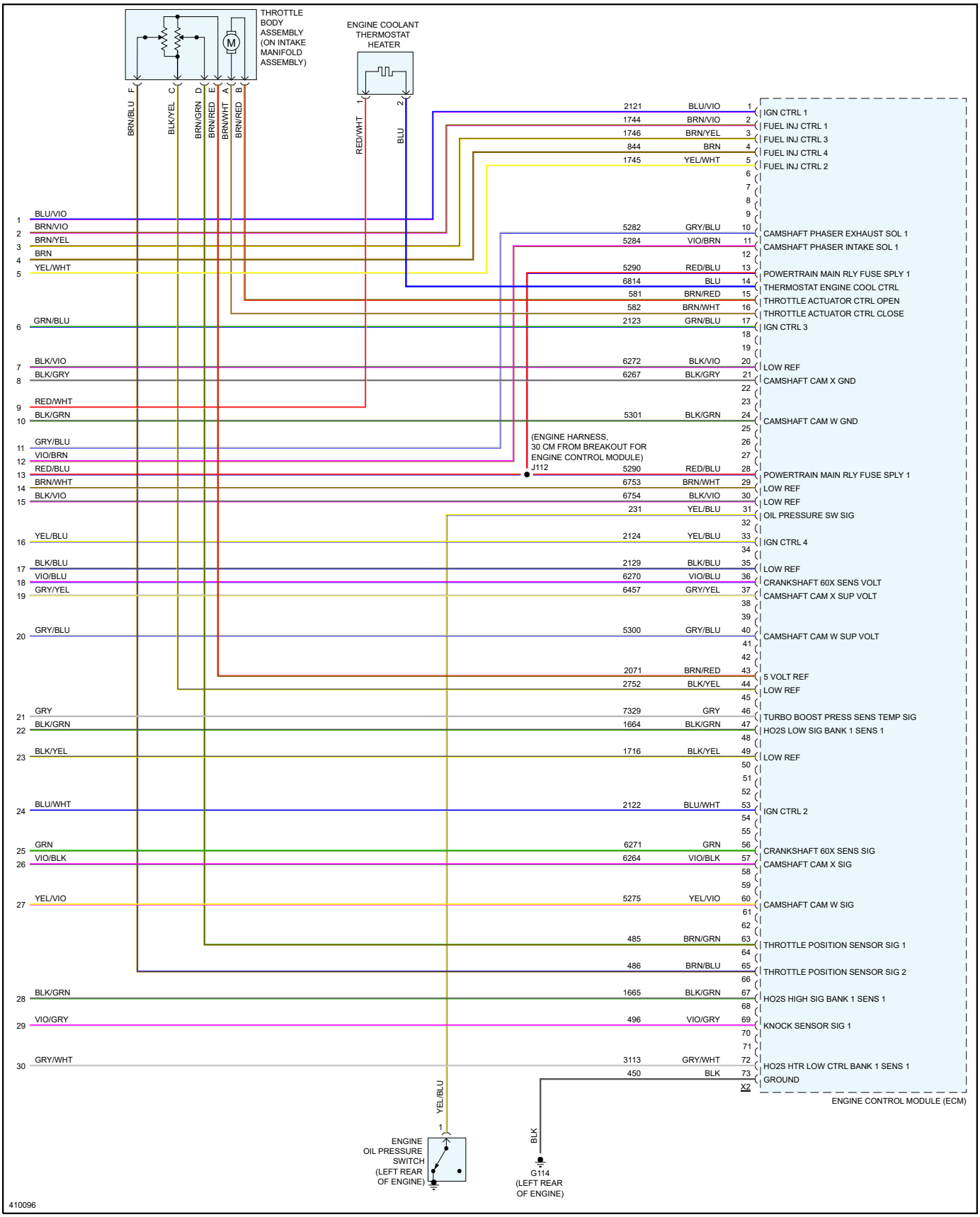

(TOP OF TRANSAXLE) (M/T) VEHICLE SPEED SENSOR
(TOP OF CLUTCH PEDAL) (M/T) CLUTCH PEDAL POSITION SENSOR
(TOP OF FUEL TANK) FUEL TANK PRESSURE (FTP) SENSOR
(RIGHT FRONT OF ENGINE) TURBOCHARGER BYPASS SOLENOID VALVE
INSTRUMENT CLUSTER
LOGIC
LO SPD GMLAN SERIAL DATA
IGN
MAL-FUNCTION IND
ENGINE OIL PRESSURE IND
COMPUTER DATA LINES SYSTEM
REVERSE SW SIG
EXTERIOR LIGHTS SYSTEM
EVAP CANISTER PUR SOL CTRL
OVERRUN AIR VALVE SOL
WASTE GATE SOL CTRL
GENERATOR FIELD DUTY CYCLE SIG
STARTING/CHARGING SYSTEM
LOW REF
(ENGINE HARNESS, 9 CM FROM BREAKOUT FOR ENGINE CONTROL MODULE) J127
INSTRUMENT CLUSTER SYSTEM
5 VOLT REF
HO2S HTR HI CTRL BANK1 SENS2
5VOLT REF
CLUTCH APPLY SENS VOLT REF
MANIFOLD ABSOLT PRESS SENS SIG
ENGINE COOLANT TEMP SENS SIG
OUTPUT SPD DIGITAL SIG
CLH APP SENS SIG
BRK POS SENS SIG
FUEL TANK PRESS SENS SIG
PRIMARY FUEL LVL SENS SIG
HO2S HIGH SIG BANK 1 SENS 2
HO2S LOW SIG BANK 1 SENS 2
ENGINE CONTROL MODULE (ECM)
MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR (IN INTAKE MANIFOLD)
BRAKE PEDAL POSITION SENSOR (TOP OF BRAKE PEDAL ASSEMBLY)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 1 (REAR OF ENGINE)
TURBOCHARGER WASTEGATE SOLENOID VALVE (LEFT SIDE OF ENGINE)
M/T
A/T
410091


(EXHAUST, DOWNSTREAM OF CATALYTIC CONVERTER) HEATED OXYGEN SENSOR 2
(EXHAUST MANIFOLD) HEATED OXYGEN SENSOR 1
(INTAKE MANIFOLD ASSEMBLY) EVAPORATIVE EMISSION (EVAP) PURGE SOLENOID VALVE
(RIGHT SIDE OF FUEL TANK) EVAPORATIVE EMISSION VENT SOLENOID VALVE
NCA
BRN/GRY
GRN
BLK/GRN
RED/WHT
GRY/WHT
GRN/BLU
RED/WHT
WHT
RED/BLU
BLU
X401
X190
RED/WHT
BLU/YEL
GRY/BLU
VIO/BRN
BRN/WHT
BLK/VIO
BLK/GRN
VIO/BLU
WHT
BRN/GRY
BLK/BRN
WHT/RED
YEL/WHT
YEL/BLK
BRN/VIO
BLK/BLU
GRN/WHT
GRN
UNDERHOOD FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
X3
X1
X104
GRN/VIO
VIO/BRN
BRN/WHT
YEL/WHT
WHT/RED
BRN/VIO
VIO/BLU
BLK
BRN/RED
DK GRN/WHT
LT GRN/WHT
X200
EXHAUST CAMSHAFT POSITION (CMP) ACTUATOR SOLENOID VALVE (LEFT FRONT OF ENGINE)
INTAKE CAMSHAFT POSITION (CMP) ACTUATOR SOLENOID VALVE (RIGHT FRONT OF ENGINE)
ENGINE COOLANT TEMPERATURE (ECT) SENSOR 2 (RIGHT SIDE OF RADIATOR)
MULTIFUNCTION INTAKE AIR SENSOR
G114 (LEFT REAR OF ENGINE)
ACCELERATOR PEDAL POSITION (APP) SENSOR (NEAR ACCELERATOR PEDAL BRACKET)
410092


(BEHIND FRONT OF GRILLE)
(W/ ENGINE IMPROVEMENT FUEL ECONOMY)
ACTIVE GRILLE AIR SHUTTER ACTUATOR
LOGIC
GND
IGN
SIG
HOT AT ALL TIMES
HOT W/ IGN MAIN RELAY ENERGIZED
HOT AT ALL TIMES
UNDERHOOD FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
ENGINE CONTROL IGNITION RELAY
FUSE 4 10A
FUSE 5 15A
FUSE 16 7.5A
FUSE 18 10A
FUSE 2 15A
FUSE 47 10A
FUSE 11 10A
FUSE 8 15A
FUSE 10 15A
FUSE 67 20A
FUSE 9 15A
(NOT USED)
G101 (LEFT FRONT CORNER OF ENGINE COMPT)
J144 (A/T)
J111
J114
X100
X190
ENGINE CONTROL MODULE (ECM)
ACC PEDAL POS SIG 2
MAS AIR FLOW SENS SIG
PRESS SENS SIG
ACC PEDAL POS SIG 1
PRE TR AIR TEMP PR SIG
A/C RFGT PR SENS SIG
LOW REF
5 VOLT REF
WAKEUP SERIAL DATA
RUN/CRANK IGN 1 VOLT
BATTERY POSITIVE VOLT
PRES SENS 5 V REF
5 VOLT REF
5 VOLT REF
5 VOLT REF
STARTR ENBLE RLY CTRL
PRIMARY RLY CTRL
HS GMLAN +
LOW REF
LOW REF
LOW REF
CRUISE/ETC/TCC BRK SIG
PARK NETRL POS SW SIG
LOW REF
LOW REF
POWERTRAIN RLY CTRL
LOW SPD COOLING FAN RLY CTRL
HS GMLAN -
TEMP SENS SIG
TEMP SENS 2 SIG
CHARGE IND CTRL
CHECK ENGINE IND CTRL
HIGH SPEED RELAY CTRL
EVAP CANSTR VENT CTRL
COMPUTER DATA LINES SYSTEM
STARTING/CHARGING SYSTEM
COMPUTER DATA LINES SYSTEM
EXTERIOR LIGHTS SYSTEM
TRANSMISSIONS SYSTEM
COOLING FANS SYSTEM
COMPUTER DATA LINES SYSTEM
STARTING/CHARGING SYSTEM
COOLING FANS SYSTEM
410093


(TOP OF VALVE COVER)
IGNITION COIL
HOT AT ALL TIMES
BATTERY FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
FUSE 5 250A
X2
IGN
GROUND
IGN CTRL 2
LOW REF
IGN CTRL 4
IGN CTRL 3
IGN CTRL 1
LOGIC
(RIGHT SIDE OF ENGINE)
FUEL INJECTORS
G112
(BODY HARNESS, 9 CM FROM RIGHT REAR WHEEL SPEED SENSOR BREAKOUT)
J312
W/ ENGINE IMPROVEMENT FUEL ECONOMY
COMPUTER DATA LINES SYSTEM
J404
(BODY HARNESS, 13 CM FROM BREAKOUT FOR FUEL PUMP CONTROL MODULE)
G402
(RIGHT SIDE OF LUGGAGE COMPT)
NCA
(SHIELD BLUNT CUT)
INSTRUMENT CLUSTER SYSTEM
X190
X401
INTAKE MANIFOLD PRESSURE/AIR TEMPERATURE SENSOR (IN AIR INTAKE DUCT, REAR OF ENGINE)
FUEL PUMP CONTROL MODULE (RIGHT SIDE OF LUGGAGE COMPT)
FUEL PUMP & LEVEL SENSOR ASSEMBLY (TOP OF FUEL TANK ASSEMBLY)
FUEL PUMP
410094


INTAKE CAMSHAFT POSITION SENSOR (RIGHT FRONT OF CYLINDER HEAD)
FUEL PRESSURE SENSOR (FUEL LINE AT RIGHT FRONT OF FUEL TANK)
EXHAUST CAMSHAFT POSITION SENSOR (LEFT FRONT OF CYLINDER HEAD)
CRANKSHAFT POSITION SENSOR (LOWER RIGHT REAR OF ENGINE)
YEL/VIO 3
GRY/BLU 1
BLK/GRN 2
BLU/BRN 1
BRN/RED 3
BLK/YEL 2
K
H
L
X401
VIO/BLK 3
GRY/YEL 1
BLK/GRY 2
GRN 3
VIO/BLU 1
BLK/VIO 2
BLU/VIO
BRN/VIO
BRN/YEL
BRN
YEL/WHT
GRN/BLU
BLK/VIO
BLK/GRY
RED/WHT
BLK/GRN
BLU/YEL
GRY/BLU
VIO/BRN
RED/BLU
BRN/WHT
BLK/VIO
BLU/WHT
BRN/RED
YEL/BLU
BLK/BLU
BLK/GRN
BLK/YEL
GRY/WHT
GRN
WHT
BLK/BRN
BRN/RED
BLK/BRN
GRY
BLK/GRN
BLU/WHT
BRN/RED
BLK/BRN
GRN
38
39
32
X3
UNDERHOOD FUSE BLOCK (ENGINE COMPT, NEXT TO LEFT STRUT TOWER)
53
55
57
X1
BRN/RED
BLK/BRN
GRN
2
3
1
X104
BRN/RED
BLK/BRN
GRN
2
1
3
A/C REFRIGERANT PRESSURE SENSOR (RIGHT RADIATOR SUPPORT)
COMPUTER DATA LINES SYSTEM
BLK/BRN 5360
WHT 5359
BLU/YEL 5361
WHT 2501
BLU 2500
19
13
7
23
24
X1
X2
BRK APY SENS LOW REF
BRK APY SENS SIG SPLY VOLT
BRK APY SENS SIG
HS GM-LAN -
HS GM-LAN +
BODY CONTROL MODULE (UNDER CENTER DASH)
BLK/YEL
VIO/GRY
2
1
KNOCK SENSOR (RIGHT REAR OF ENGINE BLOCK)
410095



		2121	BLU/VIO	1	IGN CTRL 1
		1744	BRN/VIO	2	FUEL INJ CTRL 1
		1746	BRN/YEL	3	FUEL INJ CTRL 3
		844	BRN	4	FUEL INJ CTRL 4
		1745	YEL/WHT	5	FUEL INJ CTRL 2
				6	
				7	
				8	
				9	
1	BLU/VIO				
2	BRN/VIO	5282	GRY/BLU	10	CAMSHAFT PHASER EXHAUST SOL 1
3	BRN/YEL	5284	VIO/BRN	11	CAMSHAFT PHASER INTAKE SOL 1
4	BRN			12	
5	YEL/WHT	5290	RED/BLU	13	POWERTRAIN MAIN RLY FUSE SPLY 1
		6814	BLU	14	THERMOSTAT ENGINE COOL CTRL
		581	BRN/RED	15	THROTTLE ACTUATOR CTRL OPEN
		582	BRN/WHT	16	THROTTLE ACTUATOR CTRL CLOSE
6	GRN/BLU	2123	GRN/BLU	17	IGN CTRL 3
				18	
				19	
7	BLK/VIO	6272	BLK/VIO	20	LOW REF
8	BLK/GRY	6267	BLK/GRY	21	CAMSHAFT CAM X GND
				22	
9	RED/WHT			23	
10	BLK/GRN	5301	BLK/GRN	24	CAMSHAFT CAM W GND
				25	
11	GRY/BLU			26	
12	VIO/BRN			27	
13	RED/BLU	5290	RED/BLU	28	POWERTRAIN MAIN RLY FUSE SPLY 1
14	BRN/WHT	6753	BRN/WHT	29	LOW REF
15	BLK/VIO	6754	BLK/VIO	30	LOW REF
		231	YEL/BLU	31	OIL PRESSURE SW SIG
				32	
16	YEL/BLU	2124	YEL/BLU	33	IGN CTRL 4
				34	
17	BLK/BLU	2129	BLK/BLU	35	LOW REF
18	VIO/BLU	6270	VIO/BLU	36	CRANKSHAFT 60X SENS VOLT
19	GRY/YEL	6457	GRY/YEL	37	CAMSHAFT CAM X SUP VOLT
				38	
				39	
20	GRY/BLU	5300	GRY/BLU	40	CAMSHAFT CAM W SUP VOLT
				41	
				42	
		2071	BRN/RED	43	5 VOLT REF
		2752	BLK/YEL	44	LOW REF
				45	
21	GRY	7329	GRY	46	TURBO BOOST PRESS SENS TEMP SIG
22	BLK/GRN	1664	BLK/GRN	47	HO2S LOW SIG BANK 1 SENS 1
				48	
23	BLK/YEL	1716	BLK/YEL	49	LOW REF
				50	
				51	
				52	
24	BLU/WHT	2122	BLU/WHT	53	IGN CTRL 2
				54	
				55	
25	GRN	6271	GRN	56	CRANKSHAFT 60X SENS SIG
26	VIO/BLK	6264	VIO/BLK	57	CAMSHAFT CAM X SIG
				58	
				59	
27	YEL/VIO	5275	YEL/VIO	60	CAMSHAFT CAM W SIG
				61	
				62	
		485	BRN/GRN	63	THROTTLE POSITION SENSOR SIG 1
				64	
		486	BRN/BLU	65	THROTTLE POSITION SENSOR SIG 2
				66	
28	BLK/GRN	1665	BLK/GRN	67	HO2S HIGH SIG BANK 1 SENS 1
				68	
29	VIO/GRY	496	VIO/GRY	69	KNOCK SENSOR SIG 1
				70	
				71	
30	GRY/WHT	3113	GRY/WHT	72	HO2S HTR LOW CTRL BANK 1 SENS 1
		450	BLK	73	GROUND
				X2	